

Incremental Cost of Air Purifiers

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Subject: Incremental Cost of ENERGY STAR v2 Air Purifiers, Matched Pair Analysis

The [ENERGY STAR v2 designation for air purifiers](#) went into effect on October 17, 2020. To achieve the recognition, products must have a minimum ratio of smoke CADR (Clean Air Deliver Rate) in cubic feet per minute to power draw in watts, with different minimums based on the product's CADR:

Table 1: Minimum Smoke CADR/W Requirement

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq \text{CADR} < 100$	1.9
$100 \leq \text{CADR} < 150$	2.4
$\text{CADR} \geq 150$	2.9

During the specification revision process, EPA chose to allow reduced smoke CADR/W for low-CADR products because of evidence that low-CADR products less frequently had high CADR/W ratios and might have a steeper tradeoff between cost-effectiveness and efficiency than high-CADR products. The matched pairs analysis below supports the idea that the incremental cost is higher (both as a percentage of typical product cost and in absolute terms) for low-CADR products and provides a rough estimate of this cost.

To find matched pairs of product listings, Energy Solutions considered products sold through the online catalogs for Lowe's, Amazon, Home Depot, Target, and Walmart. Starting with the product listings these retailers designated as their top-selling models, we found matching listings according to these criteria:

- CADR difference of less than 50 for a high-quality match and less than 75 for a low-quality match
- Brand and retailer match for a high-quality match; retailer matches for a low-quality match
- Matches should include one ENERGY STAR product and one non-ENERGY STAR product

The list of models and the matched pairs are shown in the appendix.

After selecting these pairs, Energy Solutions determined the price differences between the qualifying and non-qualifying model in each pair (presented in the appendix). Additionally, each pairing was assessed as either a high-quality match or a low-quality match. In determining the average price difference, high-quality matches are all given an equal weighting of 1, while low-quality matches are assigned an equal or lower weighting in sensitivity analysis variants.

The sensitivity analysis considers weights of 0 (fully excluding the low-quality matches), 0.5 (giving low-quality matches half as much weight as the high-quality matches), and 1 (giving all matches equal weightings). The analysis also considers air purifier CADR as an important factor in the incremental cost, with low-CADR and high-CADR¹ models assessed separately. The resulting incremental costs of meeting ENERGY STAR v2 for air purifiers are as follows:

SENSITIVITY WEIGHT	ALL MODELS	HIGH-CADR ONLY	LOW-CADR ONLY
0 (EXCLUDE LOW-QUALITY)	\$25.08	\$18.04	\$28.21
0.5 (HALF WEIGHT)	\$16.75	\$5.98	\$26.44
1 (EQUAL WEIGHTS)	\$12.42	\$2.54	\$24.99

In total, there are 41 models included in the study, forming 25 matched pairs. 22 models and 14 pairs are high-CADR, and the remaining 19 models and 11 pairs are low-CADR. For use cases where the number of sales in each CADR bin are known, we recommend using the incremental cost values for sales in each bin rather than applying the overall model-weighted average to sales across all bins.

There were only 4 high-quality matches in the high-CADR category and 9 high-quality matches in the low-CADR category, so we would recommend using the half-weight incremental costs of **\$5.98 for the high-CADR category** and **\$26.44 for the low-CADR category**.

If low-quality matches are given the same weightings as high-quality matches, the average CADR of ENERGY STAR products is 15 CFM greater than that of non-listed products. If they are excluded, the CADR difference is 11 CFM. This magnitude of CADR difference is not likely to have a significant impact on the assessed incremental cost. If it does impact the incremental cost, the effect would be to overstate the incremental cost because increased CADR tends to increase product price, so the incremental cost values shown above can be seen as conservatively high estimates of the additional costs a consumer sees when purchasing an ENERGY STAR air purifier.

¹ Models in EPA's CADR bin for the largest models (150+ CFM) are considered high-CADR, while all other models are considered low-CADR.

Appendix: Models and Pairs Included

Models included:

Model	ESTAR	Retailer	Smoke CADR (CFM)	2021 Price (\$)	Top Seller for Retailer?	Brand
HPA300	Yes	Lowe's	300	\$249.99	Yes	Honeywell
HPA200	Yes	Lowe's	200	\$199.99	No	Honeywell
FAP-TO3-A2	No	Lowe's	240	\$236.98	No	Filtrete
OSAP5BW1	No	Lowe's	160	\$139.99	Yes	OdorStop
TP01	No	Lowe's	70	\$399.00	No	Dyson
TP04	Yes	Lowe's	81	\$449.99	Yes	Dyson
FAP-CO3-A2	No	Lowe's	158	\$176.98	Yes	Filtrete
FP-K50UW	Yes	Lowe's	167	\$229.99	No	Sharp
5500-2	Yes	Lowe's	232	\$149.99	No	Winix
5500-2	Yes	Amazon	232	\$157.60	Yes	Winix
LV-H132	No	Amazon	80	\$76.40	Yes	Levoit
LV-PUR131S	Yes	Amazon	86	\$156.20	No	Levoit
Core 300	Yes	Amazon	140	\$99.99	Yes	Levoit
Vital 100	No	Amazon	130	\$107.98	No	Levoit
BS-03	No	Amazon	70	\$49.99	Yes	Partu
HPA020B	Yes	Amazon	53	\$62.60	No	Honeywell
AC4825E	Yes	Amazon	99	\$84.82	Yes	Germ Guardian
50250-S	No	Amazon	250	\$219.99	No	Honeywell
HPA3300BV1	Yes	Home Depot	342	\$269.00	Yes	Honeywell
BreatheSmart Classic	No	Home Depot	300	\$649.00	Yes	Alen

D480	Yes	Home Depot	310	\$219.00	Yes	Winix
5500-2	Yes	Home Depot	232	\$157.60	Yes	Winix
HPA5300B	No	Home Depot	323	\$271.99	No	Honeywell
75i-Pure	Yes	Home Depot	331	\$749.00	No	Alen
LP350TH	No	Home Depot	203	\$169.95	No	LivePure
FAP-TT-ADH	No	Walmart	50	\$38.88	Yes	Filtrete
5500-2	Yes	Walmart	232	\$156.12	Yes	Winix
HPA300	Yes	Walmart	300	\$209.00	Yes	Honeywell
AP-1512HH	Yes	Walmart	234	\$199.00	Yes	Coway
C535	No	Walmart	232	\$149.98	No	Winix
FAP-C01-F1	Yes	Walmart	68	\$49.94	No	Filtrete
50250-S	No	Walmart	250	\$219.99	No	Honeywell
200M	No	Walmart	233	\$229.98	No	Coway
HAP9425B	No	Target	130	\$69.99	Yes	Holmes
AC4825	Yes	Target	99	\$89.99	Yes	Germ Guardian
HAP9413	No	Target	75	\$49.99	Yes	Holmes
TP01	No	Target	70	\$399.99	Yes	Dyson
HAP360	Yes	Target	106	\$79.99	No	Holmes
AC4150	No	Target	66	\$69.99	No	Germ Guardian
AC4200	Yes	Target	68	\$79.99	No	Germ Guardian
DP04	Yes	Target	82	\$449.99	No	Dyson

Matched pairs:

ES v2 Model (Retailer)	Non-ES Model (Ret.)	Price Difference	CADR Difference	Match Quality	CADR Bin
TP04 (L)	TP01 (L)	\$50.99	11	High	Low-CADR
HPA300 (L)	FAP-TO3-A2 (L)	\$13.01	60	Low	High-CADR
HPA200 (L)	FAP-TO3-A2 (L)	-\$36.99	-40	Low	High-CADR

FP-K50UW (L)	OSAP5BW1 (L)	\$90.00	7	Low	High-CADR
5500-2 (L)	OSAP5BW1 (L)	\$10.00	72	Low	High-CADR
FP-K50UW (L)	FAP-CO3-A2 (L)	\$53.01	9	Low	High-CADR
5500-2 (L)	FAP-CO3-A2 (L)	-\$26.99	74	Low	High-CADR
AC4825E (A)	LV-H132 (A)	\$8.42	19	Low	Low-CADR
Core 300 (A)	Vital 100 (A)	-\$7.99	10	High	Low-CADR
LV-PUR131S (A)	LV-H132 (A)	\$79.80	6	High	Low-CADR
HPA020B (A)	BS-03 (A)	\$12.61	-17	Low	Low-CADR
5500-2 (A)	50250-S (A)	-\$62.39	-18	Low	High-CADR
HPA3300BV1 (H)	HPA5300B (H)	-\$2.99	19	High	High-CADR
75i-Pure (H)	BreatheSmart Classic (H)	\$100.00	31	High	High-CADR
D480 (H)	HPA5300B (H)	-\$52.99	-13	Low	High-CADR
FAP-C01-F1 (W)	FAP-TT-ADH (W)	\$11.06	18	High	Low-CADR
5500-2 (W)	C535 (W)	\$6.14	0	High	High-CADR
HPA300 (W)	50250-S (W)	-\$10.99	50	Low	High-CADR
AP-1512HH (W)	200M (W)	-\$30.98	1	High	High-CADR
HAP360 (T)	HAP9425B (T)	\$10.00	-24	High	Low-CADR
HAP360 (T)	HAP9413 (T)	\$30.00	31	High	Low-CADR
AC4825 (T)	AC4150 (T)	\$20.00	33	High	Low-CADR
AC4200 (T)	AC4150 (T)	\$10.00	2	High	Low-CADR
DP04 (T)	TP01 (T)	\$50.00	12	High	Low-CADR
5500-2 (H)	LP350TH (H)	-\$12.35	29	Low	High-CADR